2026 Chantilly Math Competition

Elementary School Division - Team Round

6TH, 7TH, AND 8TH GRADERS **MUST** COMPETE IN THE MIDDLE SCHOOL DIVISION!

Competition Instructions

1. Do not open the test until instructed.
2. No computational aids other than pencil/pen are permitted.
3. Write answers in designated boxes on answer sheet.
4. All answers must be nonnegative integers: $\{0, 1, 2, \ldots\}$.
5. Points per question range from 9 to 16.
6. The test consists of 16 short-answer questions to be solved in 60 minutes.
7. The final question will be used for tie-breaking.
8. There is no penalty for incorrect answers.
9. You will be graded on your answers—not your work.
10. If you have any questions, ask your proctor.
11. **Notation/Definitions:**
    - The dot $\cdot$ denotes multiplication: $a \cdot b = a \times b$.
    - $n!$ denotes the factorial of n: $n! = n \times (n-1) \times \cdots \times 1$.
    - Two integers a and b are *relatively prime* if and only if $\gcd(a, b) = 1$.

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1. (9 pts) Given that
$$f(x) = 7x + 5,$$
compute $f(5)$.

2. (9 pts) 2 trains are on the same track traveling towards each other. 1 train travels at a speed of 11 mph and the other travels at a speed of 12 mph. When the 2 trains are 2024 miles apart, a bird begins to fly back and forth between the 2 trains at a constant speed of 13 mph. How many miles will the bird travel before the trains meet?

3. (10 pts) An equilateral triangle and regular hexagon have the same area. The regular hexagon has side length 1. Let p_t and p_h be the perimeter of the equilateral triangle and the perimeter of the regular hexagon, respectively. Compute $p_t^2 + p_h^2$.

4. (10 pts) Given $xy + yz = 9$, $yz + zx = 21$, and $zx + xy = 24$, compute $|xyz|$.

5. (11 pts) A number is a super-prime if it is both prime and has the sum of its digits also prime. What is the smallest super-prime that has the sum of its digits not be a super-prime?

6. (11 pts) Moe has an empty multiplication table with 9 rows and 9 columns. Moe fills every cell of the table by writing xy in the cell in the xth row and the yth column. What is the sum of all 81 numbers Moe writes?

7. (12 pts) What 2-digit positive integer do you get when you double the answer to this problem, reverse its digits, and add 1?

8. (12 pts) A rhombus with perimeter 100 rests flat on one of its sides. The height of the rhombus is 24. Compute the sum of the lengths of the rhombus' 2 diagonals.

9. (13 pts) John wants to buy n cheeseballs. He can only buy cheeseballs in packs of 7 or 4. For how many positive integer values of n can John NOT buy EXACTLY n cheeseballs?

10. (13 pts) A quadrilateral has its vertices at coordinates $(-5, 0)$, $(-3, 4)$, $(3, -4)$, and (a, b). The 4 sides have midpoints of $(-4, 2)$, $(-1, -2)$, $(8, 7)$, $(11, 3)$ in some order. Compute $6a+7b$.

11. (14 pts) Arsenii, Justin, Rishi, and 6 other Math Club members line up for a group photo. Arsenii, Justin, and Rishi act silly whenever any 2 of them are adjacent in the lineup. How many ways are there to line up all 9 Math Club members such that Arsenii, Justin, and Rishi all don't act silly?

12. (14 pts) Let $S(n)$ denote the sum of the digits of a positive integer n.

    What is the smallest positive integer x such that 13 divides both $S(x)$ and $S(x + 1)$?

13. (15 pts) There is an empty 3×3 grid. In every square, a distinct positive integer divisor of 36 is written. The product of the 3 numbers along any row, column, or diagonal of the grid is always the same. Compute the sum of the numbers in the 4 corners of the grid.

14. (15 pts) An ant is on a vertex of a cube. Every second, it moves to a randomly chosen adjacent vertex. What integer is nearest to 100 times the probability that the ant visits the vertex of the cube farthest from its starting point before returning to its starting point for the first time?

15. (16 pts) There exists a prime p and positive integers a, b, and c, such that

$$\frac{pa+1}{2}+\frac{pb+2}{3}-\frac{pc+3}{5}$$

is an integer divisible by p. Compute the smallest possible value of $p+a+b+c$.

16. (16 pts) Carl has a triangular piece of cheese. Whenever Carl cuts a piece of cheese, he cuts it along a straight line, splitting it into 2 pieces of cheese. Carl always cuts one piece at a time. After several cuts, Carl has 21 pieces of cheese, and the total number of sides of all these pieces of cheese is 67. Every cut passed through 0, 1, or 2 existing vertices of the piece of cheese it cut. What is the total number of vertices passed through?

TB. (Tiebreak) Ryan has a random number generator with 2 integer parameters $1=m<M=67$. While $M-m>1$, the random number generator picks some integer between m and M exclusive, and Ryan changes either m or M, each option with probability $\frac{1}{2}$, to that integer. Estimate the integer nearest to 1 billion times the probability that $m=41$ when $M-m=1$.

2026 Chantilly Math Competition - Answer Sheet
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Team Name: ______________________________

Team Member 1: ______________________________

Team Member 2: ______________________________

Team Member 3: ______________________________

Team Member 4: ______________________________

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16

TB